

Answers - Log problems (page ??)

1. By taking $\log_3$ of both sides we can form a quadratic:

$$\log_3(x^{\log_3(x)}) = \log_3(81x^3)$$

$$\log_3(x) \times \log_3(x) = \log_3(81) + \log_3(x^3)$$

$$(\log_3(x))^2 = 4 + 3\log_3(x)$$

Using the substitution $u = \log_3(x)$:

$$u^2 - 3u - 4 = 0$$

$$u = -1, 4$$

Solving:

$$\log_3(x) = -1 \rightarrow x = 3^{-1} = \frac{1}{3}$$

$$\log_3(x) = 4 \rightarrow x = 3^4 = 81$$

2. $4^{x-1} = 2^x + 48$

$$(2^2)^{x-1} = 2^x + 48$$

$$2^{2x-2} = 2^x + 48$$

$$\frac{2^{2x}}{2^2} = 2^x + 48$$

$$\frac{(2^x)^2}{4} = 2^x + 48$$

Substitute $u = 2^x$ and solve the quadratic:

$$\frac{u^2}{4} - u - 48 = 0$$

$$u = 16, -12$$

Reverse substitution:

$$2^x = 16 \rightarrow x = 4$$

$$2^x = -12 \rightarrow \text{Not possible}$$

Therefore the only solution is $x = 4$

3. Use the change of base formula to change to base 10:

$$\frac{\log(y)}{\log(x)} + \frac{\log(x)}{\log(y)} = 2$$

$$\frac{(\log(y))^2 + (\log(x))^2}{\log(x)\log(y)} = 2$$

$$(\log(y))^2 + (\log(x))^2 = 2\log(x)\log(y)$$

$$(\log(y))^2 - 2\log(x)\log(y) + (\log(x))^2 = 0$$

This is a perfect square, so factorise:

$$(\log(x) - \log(y))^2 = 0$$

$$\log(x) = \log(y)$$

$$x = y$$

$$\frac{x}{y} + \frac{y}{x} = \frac{x}{x} + \frac{x}{x} = 2$$

4. If $\sqrt{\log_a(b)} + \sqrt{\log_b(a)} = 2$, then find the value of $\log_{ab}(a) - \log_{\frac{1}{ab}}(b)$

Squaring the equation gives us:

$$\log_a(b) + 2\sqrt{\log_a(b)\log_b(a)} + \log_b(a) = 4$$

Use change of base formula to simplify:

$$\frac{\log(b)}{\log(a)} + 2\sqrt{\frac{\log(b)}{\log(a)} \times \frac{\log(a)}{\log(b)}} + \frac{\log(a)}{\log(b)} = 4$$

$$\frac{\log(b)}{\log(a)} + 2\sqrt{1} + \frac{\log(a)}{\log(b)} = 4$$

$$\frac{\log(b)}{\log(a)} + 2 + \frac{\log(a)}{\log(b)} = 4$$

$$\frac{\log(b)}{\log(a)} + \frac{\log(a)}{\log(b)} = 2$$

$$\frac{(\log(b))^2 + (\log(a))^2}{\log(a)\log(b)} = 2$$

$$(\log(b))^2 + (\log(a))^2 = 2\log(a)\log(b)$$

$$(\log(b))^2 - 2\log(a)\log(b) + (\log(a))^2 = 0$$

Perfect square:

$$(\log(b) - \log(a))^2 = 0$$

$$\log(a) = \log(b)$$

$$a = b$$

Substituting into $\log_{ab}(a) - \log_{\frac{1}{ab}}(b)$ we get $\log_{a^2}(a) - \log_{\frac{1}{a^2}}(a)$

$$\log_{a^2}(a) = \frac{1}{2} \text{ and } \log_{\frac{1}{a^2}}(a) = \frac{-1}{2}$$

$$\frac{1}{2} - \frac{-1}{2} = 1$$

5. If $2^{3x-5} = 3^{x+3}$ and $x = \log(864^{\log_{10}(y)})$, then find the value of $y^{\log_{10} \frac{8}{3}}$

Taking log base 10 of both sides:

$$(3x - 5)\log_{10}(2) = (x + 3)\log_{10}(3)$$

$$3x\log_{10}(2) - 5\log_{10}(2) = x\log_{10}(3) + 3\log_{10}(3)$$

$$3x\log_{10}(2) - x\log_{10}(3) = 5\log_{10}(2) + 3\log_{10}(3)$$

$$x(3\log_{10}(2) - \log_{10}(3)) = 5\log_{10}(2) + 3\log_{10}(3)$$

$$x = \frac{5\log_{10}(2) + 3\log_{10}(3)}{3\log_{10}(2) - \log_{10}(3)}$$

Simplify using log rules:

$$x = \frac{\log_{10}(32) + \log_{10}(27)}{\log_{10}(8) - \log_{10}(3)}$$

$$x = \frac{\log_{10}(864)}{\log_{10}(\frac{8}{3})}$$

$$x = \frac{1}{\log_{10}(\frac{8}{3})} \times \log_{10}(864)$$

$$x = \log_{10}(864)^{\frac{1}{\log_{10}(\frac{8}{3})}}$$

Going back to the original question, this means that $\log_{10}(y) = \frac{1}{\log_{10}(\frac{8}{3})}$

$$\log_{10}(\frac{8}{3}) \log_{10}(y) = 1$$

$$\log_{10}(y)^{\log_{10}(\frac{8}{3})} = 1$$

$$y^{\log_{10}(\frac{8}{3})} = 10$$

6. $7^0 = \log_9(x^2 + \sqrt{x+1} + 8)$

$$1 = \log_9(x^2 + \sqrt{x+1} + 8)$$

$$9^1 = x^2 + \sqrt{x+1} + 8$$

$$\sqrt{x+1} = 1 - x^2$$

Squaring the equation:

$$x+1 = 1 - 2x^2 + x^4$$

$$x^4 - 2x^2 - x = 0$$

Solving, we get $x = 0, -1, \frac{1 \pm \sqrt{5}}{2}$

Substituting back into the original equation (as we should because by squaring the equation we may have introduced false solutions), we find that $x = \frac{1 \pm \sqrt{5}}{2}$ is not valid, therefore $x = 0, -1$

7. If $\log_{16}(x) + \log_8(y) = 11$ and $\log_8(x) + \log_{16}(y) = 10$ then find the value of $\frac{y}{x^2}$

Because bases are all powers of 2, we will use the change of base formula to make the new base 2.

Equation 1: $\frac{\log_2(x)}{\log_2(16)} + \frac{\log_2(y)}{\log_2(8)} = 11$

Becomes $\frac{\log_2(x)}{4} + \frac{\log_2(y)}{3} = 11$ which we can rearrange into $3\log_2(x) + 4\log_2(y) = 132$

Equation 2: $\frac{\log_2(x)}{\log_2(8)} + \frac{\log_2(y)}{\log_2(16)} = 10$

Becomes $\frac{\log_2(x)}{3} + \frac{\log_2(y)}{4} = 10$ which we can rearrange into $3\log_2(x) + 3\log_2(y) = 120$

Solving simultaneously:

$$3\log_2(x) + 4\log_2(y) = 132$$

$$3\log_2(x) + 3\log_2(y) = 120$$

$$\log_2(y) = 24$$

$$y = 2^{24}$$

Solve for x by substituting back into equation 1:

$$3\log_2(x) + 4\log_2(2^{24}) = 132$$

$$3\log_2(x) + 4 \times 24 = 132$$

$$3\log_2(x) = 36$$

$$\log_2(x) = 12$$

$$x = 2^{12}$$

To find the value of $\frac{y}{x^2}$ we substitute:

$$\frac{y}{x^2} = \frac{2^{24}}{(2^{12})^2} = \frac{2^{24}}{2^{24}} = 1$$

8. Use the change of base formula to change the base to 2:

$$\frac{\log_2(4)}{\log_2(\log_2(x))} = \log_2\left(\frac{\log_2(x)}{\log_2(4)}\right)$$

$$\frac{2}{\log_2(\log_2(x))} = \log_2\left(\frac{\log_2(x)}{2}\right)$$

$$\frac{2}{\log_2(\log_2(x))} = \log_2(\log_2(x)) - \log_2(2)$$

$$\frac{2}{\log_2(\log_2(x))} = \log_2(\log_2(x)) - 1$$

$$2 = (\log_2(x))^2 - \log_2(\log_2(x))$$

We have a quadratic in terms of $\log_2(\log_2(x))$, so we make a substitution:

$$u = \log_2(\log_2(x))$$

$$2 = u^2 - u$$

$$u^2 - u - 2$$

$$u = 2, -1$$

Reversing the substitution:

$$\log_2(\log_2(x)) = 2$$

$$\log_2(x) = 2^2 = 4$$

$$2^4 = x \rightarrow x = 16$$

$$\log_2(\log_2(x)) = -1$$

$$\log_2(x) = 2^{-1} = \frac{1}{2}$$

$$2^{\frac{1}{2}} = x \rightarrow x = \sqrt{2}$$

$$x = \sqrt{2}, 16$$

9. Use the change of base formula for each equation, then simplify (notice we are using bases that help us get whole number bases).

$$\text{Equation 1: } \frac{\log_2(x)}{\log_2(4)} + \frac{\log_3(y)}{\log_3(9)} = 2$$

$$\frac{\log_2(x)}{2} + \frac{\log_3(y)}{2} = 2$$

$$\log_2(x) + \log_3(y) = 4$$

$$\text{Equation 2: } \frac{\log_2(2)}{\log_2(x)} + \frac{\log_3(3)}{\log_3(y)} = 1$$

$$\frac{1}{\log_2(x)} + \frac{1}{\log_3(y)} = 1$$

$$\frac{\log_2(x) + \log_3(y)}{\log_2(x) \log_3(y)} = 1$$

$$\log_2(x) + \log_3(y) = \log_2(x) \log_3(y)$$

Substitute equation 1 into equation 2:

$$4 = \log_2(x) \log_3(y)$$

Rearrange and make $\log_3(y)$ the subject:

$$\log_3(y) = \frac{4}{\log_2(x)}$$

Substitute into equation 1:

$$\log_2(x) + \frac{4}{\log_2(x)} = 4$$

This is a quadratic in terms of $\log_2(x)$, so we substitute $u = \log_2(x)$:

$$u + \frac{4}{u} = 4$$

$$u^2 - 4u + 4 = 0$$

$$u = 2$$

Reverse the substitution:

$$\log_2(x) = 2$$

$$x = 2^2 = 4$$

Substitute into $\log_3(y) = \frac{4}{\log_2(x)}$

$$\log_3(y) = \frac{4}{\log_2(4)}$$

$$\log_3(y) = 2$$

$$y = 3^2 = 9$$

10. If $\log_5(4)$, $\log_5(2^x + \frac{1}{2})$ and $\log_5(2^x - \frac{1}{4})$ are in arithmetic progression, find the value of x and also find the common difference.

$$\log_5(2^x + \frac{1}{2}) - \log_5(4) = \log_5(2^x - \frac{1}{4}) - \log_5(2^x + \frac{1}{2})$$

$$2 \log_5(2^x + \frac{1}{2}) = \log_5(4) + \log_5(2^x - \frac{1}{4})$$

$$2 \log_5(2^x + \frac{1}{2}) = \log_5 4(2^x - \frac{1}{4})$$

$$2 \log_5(2^x + \frac{1}{2}) = \log_5(4 \times 2^x - 1)$$

$$\log_5(2^x + \frac{1}{2})^2 = \log_5(4 \times 2^x - 1)$$

$$\log_5 \left(2^x + \frac{1}{2} \right)^2 - \log_5(4 \times 2^x - 1) = 0$$

$$\log_5 \left(\frac{(2^x + \frac{1}{2})^2}{4 \times 2^x - 1} \right) = 0$$

$$\frac{(2^x + \frac{1}{2})^2}{4 \times 2^x - 1} = 1$$

$$\left(2^x + \frac{1}{2} \right)^2 = 4 \times 2^x - 1$$

$$(2^x)^2 + 2^x + \frac{1}{4} = 4 \times 2^x - 1$$

$$(2^x)^2 - 3 \times 2^x + \frac{5}{4} = 0$$

Substitute $u = 2^x$:

$$u^2 - 3u + \frac{5}{4} = 0$$

$$u = \frac{5}{2}, \frac{1}{2}$$

Reverse substitution:

$$2^x = \frac{1}{2} \rightarrow x = -1$$

$$2^x = \frac{5}{2}$$

$$x = \log_2 \frac{5}{2} = \log_2(5) - 1$$

Substitute into original terms to get common difference:

$$d = \log_5(2^x - \frac{1}{4}) - \log_5(2^x + \frac{1}{2})$$

$$\log_5(\frac{1}{2} - \frac{1}{4}) - \log_5(\frac{1}{2} + \frac{1}{2}) \text{ (substituting } 2^x = \frac{1}{2})$$

$$d = \log_5(\frac{1}{4}) - \log_5(1)$$

$$d = \log_5(\frac{1}{4}) = \log_5(4^{-1}) = -\log_5(4)$$

$$\log_5(\frac{5}{2} - \frac{1}{4}) - \log_5(\frac{5}{2} + \frac{1}{2}) \text{ (substituting } 2^x = \frac{5}{2})$$

$$d = \log_5(\frac{9}{4}) - \log_5(3) = \log_5(\frac{3}{4})$$

$$d = \log_5(3) - \log_5(4)$$

11. Start with equation 2:

$$\log_{10}\left(\frac{x+y}{x-y}\right) = \log_{10}(8)$$

$$\frac{x+y}{x-y} = 8$$

$$x + y = 8x - 8y$$

$$9y = 7x$$

$$y = \frac{7x}{9}$$

Substitute into equation 1:

$$\log_{10}(x^2 + (\frac{7x}{9})^2) = 1 + \log_{10}(13)$$

$$\log_{10}\left(x^2 + (\frac{49x^2}{81})\right) = 1 + \log_{10}(13)$$

$$\log_{10}\left(\frac{130x^2}{81}\right) = \log_{10}(10) + \log_{10}(13)$$

$$\log_{10}\left(\frac{130x^2}{81}\right) = \log_{10}(130)$$

$$\frac{130x^2}{81} = 130$$

$$\frac{x^2}{81} = 1$$

$$x^2 = 81$$

$$x = \pm 9$$

Substitute into $y = \frac{7x}{9}$

$$y = 7, -7$$

Solutions are $x = 9, y = 7$ and $x = -9, y = -7$

However, we can't have a negative solution as $\log_{10}(-9 - -7)$ is undefined.

Therefore, $x = 9, y = 7$

12. Evaluate the expression:

$$\frac{1}{1+\log_a(bc)} + \frac{1}{1+\log_b(ac)} + \frac{1}{1+\log_c(ab)}$$

Change everything to base 10:

$$\begin{aligned} & \frac{1}{1+\frac{\log(bc)}{\log(a)}} + \frac{1}{1+\frac{\log(ac)}{\log(b)}} + \frac{1}{1+\frac{\log(ab)}{\log(c)}} \\ & \frac{1}{\frac{\log(a)+\log(bc)}{\log(a)}} + \frac{1}{\frac{\log(b)+\log(ac)}{\log(b)}} + \frac{1}{\frac{\log(c)+\log(ab)}{\log(c)}} \\ & \frac{\log(a)}{\log(a)+\log(bc)} + \frac{\log(b)}{\log(b)+\log(ac)} + \frac{\log(c)}{\log(c)+\log(ab)} \\ & \frac{\log(a)}{\log(a)+\log(b)+\log(c)} + \frac{\log(b)}{\log(b)+\log(a)+\log(c)} + \frac{\log(c)}{\log(c)+\log(a)+\log(b)} \\ & \frac{\log(a)+\log(b)+\log(c)}{\log(a)+\log(b)+\log(c)} = 1 \end{aligned}$$

13. Take log of base 10 of both sides.

$$\log(x^{\log(x)}) = \log\left(\frac{x^3}{100}\right)$$

$$(\log(x))(\log(x)) = \log(x^3) - \log(100)$$

$$(\log(x))^2 = 3\log(x) - 2$$

$$(\log(x))^2 - 3\log(x) + 2 = 0$$

$$\log(x) = 1, 2$$

$$x = 10, 100$$